

NovaCart Cybersecurity Risk Assessment

Executive Summary | Fictional E-Commerce Company

Assessment Overview

This portfolio project evaluates cybersecurity risks within NovaCart, a fictional e-commerce company. The assessment covers customer and payment data, the e-commerce website, employee endpoints, corporate email, cloud storage, the internal network, and privileged accounts. Twelve risk scenarios were assessed.

Methodology & Control Reference

Risk assessment: NIST SP 800-30 Rev. 1 principles were used to structure threat, vulnerability, likelihood, impact, and risk evaluation. **Control assessment:** relevant NCA NCNICC 1:2025 control areas were used to identify security gaps and recommended actions for the simulated private-sector environment.

Key Findings

Area	Key Gap / Exposure	Priority
Identity & Access	Weak access controls and MFA not enforced for privileged access	High
Application Security	Unpatched software components in the e-commerce environment	High
Cloud & Data	Misconfigured permissions and inadequate protection of sensitive data	High
Human Risk	No recurring phishing/security awareness training	High
Endpoint & Recovery	Outdated endpoint protection, encryption and backup gaps	High

Recommended Priorities

1. Strengthen identity and privileged-access controls, including MFA and least privilege.
2. Remediate vulnerable software and formalize patch management.
3. Correct cloud access permissions and strengthen protection of customer/payment data.
4. Improve endpoint protection, device encryption, backup, and recovery practices.
5. Establish recurring phishing and cybersecurity awareness training.

Risk Treatment

The selected response for the identified scenarios is primarily **risk mitigation**. Treatment actions were assigned to appropriate business or technical owners with target completion periods. Residual risk was reassessed under the assumption that the recommended controls are implemented effectively; the remaining risk should be validated again using implementation evidence.

Conclusion

The assessment indicates that NovaCart's most important improvement areas are access control, application security, data protection, endpoint security, and resilience. Implementing the recommended controls would reduce the likelihood of several major threat scenarios while improving the organization's overall cybersecurity posture.

Portfolio disclaimer: NovaCart and all assessment conditions are fictional. Risk ratings, control statuses, owners, and treatment timelines are simulated for educational and portfolio purposes.